

Python 30 Days - Master Notes

DAY 1: Python Fundamentals + Conditional Statements

Color guide: **Explanation** | **Code** | **Output** | **Important** / **Revision**

How to use these notes

Read the explanation first, then study the code, predict the output, and finally run the same example in VS Code. The notes are written for revision, so each topic includes the meaning, syntax, example, output, explanation, and important points.

1. print() - Displaying Output

print() is used to display text, numbers, or calculation results on the screen. Text is normally written inside quotes. Numbers do not need quotes.

Code

```
print("Hello World")
print("I am learning Python")
print(100)
print(10 + 20)
```

Expected output

```
Hello World
I am learning Python
100
30
```

Important to remember

Quotes make something text. **print(10 + 20)** performs the calculation first and prints 30.

2. Comments

Comments are notes for humans reading the program. Python ignores comments when the program runs. A single-line comment starts with #.

Code

```
# This is a comment
print("Python") # This is an inline comment
# print("This line will not run")
```

Expected output

```
Python
```

Important to remember

Use comments to explain **why** code exists. Do not fill every line with unnecessary comments.

3. Variables

A variable is a name used to store a value. Think of it as a labeled box. The value can be changed later by assigning a new value.

Code

```
name = "Bhanu"
age = 23
salary = 30000

print(name)
print(age)

age = 24
print(age)
```

Expected output

```
Bhanu
23
24
```

Important to remember

Use **=** to assign a value. Variable names are case-sensitive: **age** and **Age** are different.

4. Basic Data Types

A data type tells Python what kind of value it is working with. The main Day 1 types are **str**, **int**, **float**, **bool**, and **None**.

Code

```
name = "Bhanu" # str - text
age = 23 # int - whole number
price = 99.50 # float - decimal number
is_learning = True # bool - True/False
value = None # no value

print(type(name))
print(type(age))
```

Expected output

```
<class 'str'>
<class 'int'>
```

Important to remember

"10" + "20" gives **"1020"** because they are strings. **10 + 20** gives **30** because they are integers.

5. User Input

input() pauses the program and lets the user type a value. The important rule is that **input()** returns a **string**, even when the user types digits.

Code

```
name = input("Enter your name: ")
print(f"Hello {name}")
```

Expected output

```
Enter your name: Bhanu
Hello Bhanu
```

Important to remember

If you need to perform number calculations, convert the input using **int()** or **float()**.

6. Type Conversion

Type conversion changes a value from one data type to another. You practiced **int()**, **float()**, and **str()**.

Code

```
age = "23"
age = int(age)
print(age, type(age))

price = "99.50"
price = float(price)
print(price, type(price))

number = 100
number = str(number)
print(number, type(number))
```

Expected output

```
23 <class 'int'>
99.5 <class 'float'>
100 <class 'str'>
```

Important to remember

Use **int(input(...))** when you expect a whole-number input and want to compare or calculate with it.

7. Arithmetic and Assignment Operators

Operators perform operations on values. Arithmetic operators calculate numbers. Assignment operators store or update values.

Code

```
a = 10
b = 3

print(a + b) # addition
print(a - b) # subtraction
print(a * b) # multiplication
print(a / b) # division
print(a // b) # floor division
print(a % b) # remainder
print(a ** b) # power

a += 5 # same idea as a = a + 5
print(a)
```

Expected output

```
13
7
30
3.3333333333333335
3
1
1000
15
```

Important to remember

/ gives normal division, // removes the decimal part by floor division, % gives the remainder, and ** means power.

8. Strings

A string is a sequence of characters written inside quotes. You practiced indexing, slicing, length, string methods, replacement, concatenation ideas, and f-strings.

Code

```
name = "Bhanu Prakash"

print(name[0]) # first character
print(name[-1]) # last character
print(name[0:5]) # slice
print(len(name)) # length
print(name.upper())
print(name.lower())
print(name.title())

message = "I am learning Java"
print(message.replace("Java", "Python"))

age = 23
print(f"My name is {name} and I am {age} years old.")
```

Expected output

```
B
h
Bhanu
13
BHANU PRAKASH
bhanu prakash
Bhanu Prakash
I am learning Python
My name is Bhanu Prakash and I am 23 years old.
```

Important to remember

Indexing starts at **0**. **-1** means the last character. In **[start:stop]**, the stop position is excluded. Strings are immutable, so string methods normally return a new string.

9. Comparison Operators

Comparison operators compare two values. Their result is always **True** or **False**.

Code

```
age = 23

print(age == 23) # equal
print(age != 18) # not equal
print(age > 18) # greater than
print(age < 30) # less than
print(age >= 18) # greater than or equal
print(age <= 30) # less than or equal
```

Expected output

```
True
True
True
True
True
True
```

Important to remember

= means assignment. **==** means comparison. This difference is very important.

10. Conditional Statements - if / elif / else

Conditional statements allow Python to make decisions. **if** checks a condition. **elif** checks another condition when an earlier one was false. **else** runs when the previous conditions were false.

Code

```
age = int(input("Enter your age: "))

if age >= 18:
    print("You are eligible")
else:
    print("You are not eligible")

marks = int(input("Enter your marks: "))

if marks >= 90:
    print("Grade A")
elif marks >= 75:
    print("Grade B")
elif marks >= 50:
    print("Grade C")
else:
    print("Fail")
```

Expected output

```
Example 1:
Enter your age: 23
You are eligible

Example 2:
Enter your marks: 80
Grade B
```

Important to remember

Python checks an if/elif chain from **top to bottom** and stops after the first True branch. Indentation is required inside **if**, **elif**, and **else**.

11. Logical Operator - and

and combines conditions. The complete condition is True only when **both** sides are True.

Code

```
age = int(input("Enter your age: "))
python_score = int(input("Enter your Python score: "))

if age >= 18 and python_score >= 70:
    print("You are eligible")
else:
    print("You are not eligible")
```

Expected output

```
Age 23, score 88:
True and True -> True
You are eligible

Age 18, score 33:
True and False -> False
You are not eligible
```

Important to remember

and = BOTH conditions must be True.

12. Logical Operator - or

or combines conditions. The complete condition is True when **at least one** side is True.

Code

```
has_python = True
has_java = False

if has_python or has_java:
    print("You know at least one programming language")
else:
    print("You need to learn a programming language")
```

Expected output

```
You know at least one programming language
```

Important to remember

True or False is True. Only **False or False** is False.

13. Logical Operator - not

not reverses a Boolean value. **not True** becomes False, and **not False** becomes True.

Code

```
is_blocked = False

if not is_blocked:
    print("User can access the website")
else:
    print("User is blocked")
```

Expected output

```
User can access the website
```

Important to remember

Because **is_blocked** is False, **not is_blocked** becomes True.

14. Day 1 Final Combined Program

This program combines the main concepts you practiced: variables, input, type conversion, comparisons, logical operators, conditional statements, f-strings, and print().

Code

```
name = input("Enter your name: ")
age = int(input("Enter your age: "))
python_score = int(input("Enter your Python score: "))
city = input("Enter your city: ")

if age >= 18 and python_score >= 70:
    print(f"Hello {name}")
    print(f"City: {city}")
    print("You are eligible for the interview.")
else:
    print(f"Hello {name}")
    print(f"City: {city}")
    print("You are not eligible for the interview.")
```

Expected output

```
Enter your name: Bhanu
Enter your age: 23
Enter your Python score: 85
Enter your city: Bangalore
Hello Bhanu
City: Bangalore
You are eligible for the interview.
```

Important to remember

Read the condition in plain English: **If age is at least 18 AND Python score is at least 70, print eligible; otherwise print not eligible.**

Day 1 - Common Mistakes

Mistake	Why it is wrong / fix
Using = instead of ==	= assigns a value; == compares two values.
Forgetting int() for numeric input	input() returns a string. Convert it before numeric comparison/calculation.
Wrong indentation	Code inside if/elif/else must be indented consistently.
Expecting range or loops here	Loops belong to Day 2; they are not part of the Day 1 final practice.
Thinking and means either condition	and requires both conditions to be True.
Thinking or requires both	or requires at least one condition to be True.

Day 1 - Quick Revision

Item	Meaning
print()	Display output
#	Comment
variable	Stores a value
str / int / float / bool / None	Basic data types
input()	Take user input; returns a string
int() / float() / str()	Type conversion
type()	Check data type
len()	Find length
= / ==	Assignment / comparison
if / elif / else	Decision making
and	Both conditions must be True
or	At least one condition must be True
not	Reverse a Boolean

Self-check questions

1. What is the difference between = and ==?
2. What data type does **input()** return?
3. What does **and** require?
4. What does **or** require?
5. What does **not** do?
6. Why do we use **int(input(...))**?
7. What happens when an if condition is False and an else block exists?
8. Why is indentation important in Python?

Day 1 learning status: Python concepts, practice, and revision notes are complete. Git and GitHub are still remaining before Day 1 is officially marked complete.